

mable logic is applicable to the wireless side of things, especially digital front-ends. If something changes or a hash/algorithm is blacklisted or cracked, you need not replace the entire device, just update it. In this space a little re-configurability goes a long way.

Applications

Application engineers who work on software-defined radio use FPGA fabric to swap among multiple modems on the fly. That is something they cannot accomplish easily with a stand-alone custom chip. In the video business, engineers could drive various display scales and resolutions with a single device by customising the FPGA fabric to match up to various display standards.

If your application requires compute acceleration, FPGA fabric could be reconfigured to implement various custom accelerators. But if your application requires custom

hard logic that does not happen to be on those devices, EFLX will allow you to design those same benefits into your custom IC.

Enabling customers to customise and optimise

The big issue is not IP but tool chains, each of which may have a portfolio comprising hundreds and thousands of processor variations. In many cases, these variations involve relatively minor changes like pinouts. Incorporating even a single EFLX-100 core in each MCU could dramatically reduce the number of different MCU options. Also, the ability to customise the MCU's abilities using programmable fabric may have tremendous implications with regards to implementing versatile nodes for use with the Internet of Things (IoT).

As SoC designs get more complex and masks cost millions of dollars, ICs and systems companies need to

make their designs flexible enough to handle broader applications to generate more revenue and improve rate of interest, and to build the capability to upgrade critical RTL blocks to handle evolving standards and customer requirements. This flexibility enables new business models wherein system upgrades offering valuable new capabilities can be offered for a fee, or where one chip can be offered in several versions with different reconfigurable RTLs for different applications and priced to the value of each.

Architects, front-end designers and physical design teams need to become familiar with this new technology for applications from MCU to the IoT to networking and more. Evaluation and prototyping of a single-core EFLX design will give customers the experience and insight to build flexibility into their RTL designs, even if the target application requires a larger EFLX array. **EFY**

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Circuit Systems India Ltd.
B-24, GIDC Electronics Estate, Sector-25, Gandhinagar-382044, Gujarat, India
Mail: pcb@pcbpower.com | Phone: +91 7600012414/+91 7600012415

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